

Multiple Linear Regression – Multi-Channel Marketing Analysis

Load Dataset and Perform Initial EDA

```
In [36]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor

from scipy.stats import shapiro
```

```
In [37]: df = pd.read_csv("marketing_sales_data.csv")
```

Dataset Overview

```
In [38]: df.head()
```

```
Out[38]:
```

	TV	Radio	Social Media	Influencer	Sales
0	Low	3.518070	2.293790	Micro	55.261284
1	Low	7.756876	2.572287	Mega	67.574904
2	High	20.348988	1.227180	Micro	272.250108
3	Medium	20.108487	2.728374	Mega	195.102176
4	High	31.653200	7.776978	Nano	273.960377

```
In [39]: print(df.dtypes)
```

```
TV                str
Radio            float64
Social Media     float64
Influencer        str
Sales            float64
dtype: object
```

```
In [40]: df = pd.get_dummies(
    df,
    columns=["TV", "Influencer"],
    drop_first=True
)

df.head()
```

Out[40]:

	Radio	Social Media	Sales	TV_Low	TV_Medium	Influencer_Mega	Influencer_Micr
0	3.518070	2.293790	55.261284	True	False	False	Tru
1	7.756876	2.572287	67.574904	True	False	True	Fals
2	20.348988	1.227180	272.250108	False	False	False	Tru
3	20.108487	2.728374	195.102176	False	True	True	Fals
4	31.653200	7.776978	273.960377	False	False	False	Fals

In [41]:

```
df.describe()
```

Out[41]:

	Radio	Social Media	Sales
count	572.000000	572.000000	572.000000
mean	17.520616	3.333803	189.296908
std	9.290933	2.238378	89.871581
min	0.109106	0.000031	33.509810
25%	10.699556	1.585549	118.718722
50%	17.149517	3.150111	184.005362
75%	24.606396	4.730408	264.500118
max	42.271579	11.403625	357.788195

In [42]:

```
df.info()

<class 'pandas.DataFrame'>
RangeIndex: 572 entries, 0 to 571
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Radio                  572 non-null    float64
1   Social Media           572 non-null    float64
2   Sales                  572 non-null    float64
3   TV_Low                 572 non-null    bool
4   TV_Medium              572 non-null    bool
5   Influencer_Mega        572 non-null    bool
6   Influencer_Micro       572 non-null    bool
7   Influencer_Nano        572 non-null    bool
dtypes: bool(5), float64(3)
memory usage: 16.3 KB
```

In [43]:

```
# Dataset shape
print("shape:", df.shape)

# Column information
df.info()
```

```
# Summary statistics
df.describe()
```

```
shape: (572, 8)
<class 'pandas.DataFrame'>
RangeIndex: 572 entries, 0 to 571
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Radio                  572 non-null    float64
1   Social Media           572 non-null    float64
2   Sales                  572 non-null    float64
3   TV_Low                 572 non-null    bool
4   TV_Medium              572 non-null    bool
5   Influencer_Mega        572 non-null    bool
6   Influencer_Micro       572 non-null    bool
7   Influencer_Nano        572 non-null    bool
dtypes: bool(5), float64(3)
memory usage: 16.3 KB
```

```
Out[43]:
```

	Radio	Social Media	Sales
count	572.000000	572.000000	572.000000
mean	17.520616	3.333803	189.296908
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25%	10.699556	1.585549	118.718722
50%	17.149517	3.150111	184.005362
75%	24.606396	4.730408	264.500118
max	42.271579	11.403625	357.788195

Check Missing Values

```
In [44]: df.isnull().sum()
```

```
Out[44]: Radio                0
Social Media                0
Sales                      0
TV_Low                     0
TV_Medium                  0
Influencer_Mega            0
Influencer_Micro           0
Influencer_Nano            0
dtype: int64
```

```
In [45]: print(df.columns)
```

```
Index(['Radio', 'Social Media', 'Sales', 'TV_Low', 'TV_Medium',
      'Influencer_Mega', 'Influencer_Micro', 'Influencer_Nano'],
      dtype='str')
```

Explore Categories

```
In [46]: print(df["TV_Low"].value_counts())
print("\n")
print(df["TV_Medium"].value_counts())

print("\n-----\n")

print(df["Influencer_Mega"].value_counts())
print(df["Influencer_Micro"].value_counts())
print(df["Influencer_Nano"].value_counts())
```

```
TV_Low
False    371
True     201
Name: count, dtype: int64
```

```
TV_Medium
False    367
True     205
Name: count, dtype: int64
```

```
-----

Influencer_Mega
False    435
True     137
Name: count, dtype: int64
Influencer_Micro
False    421
True     151
Name: count, dtype: int64
Influencer_Nano
False    415
True     157
Name: count, dtype: int64
```

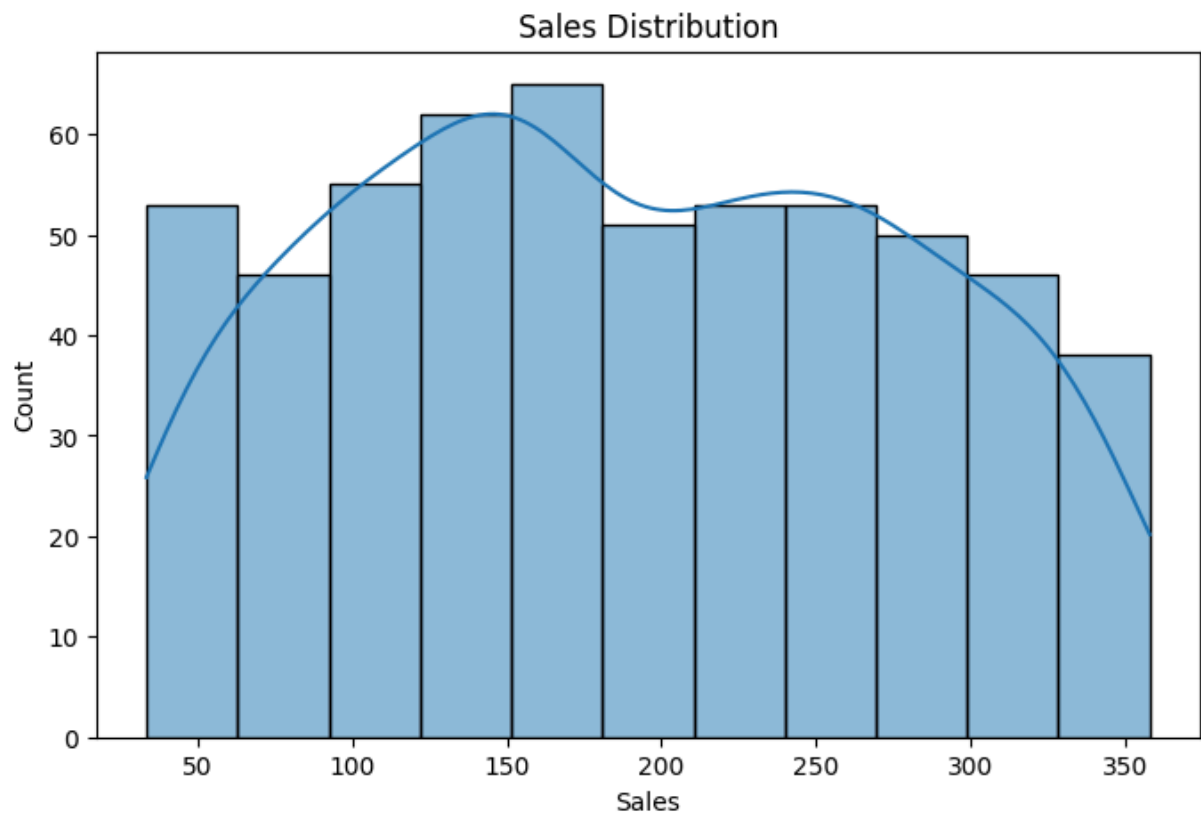
Visualize Sales Distribution

```
In [47]: import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8,5))

sns.histplot(df["Sales"], kde=True)

plt.title("Sales Distribution")
plt.show()
```

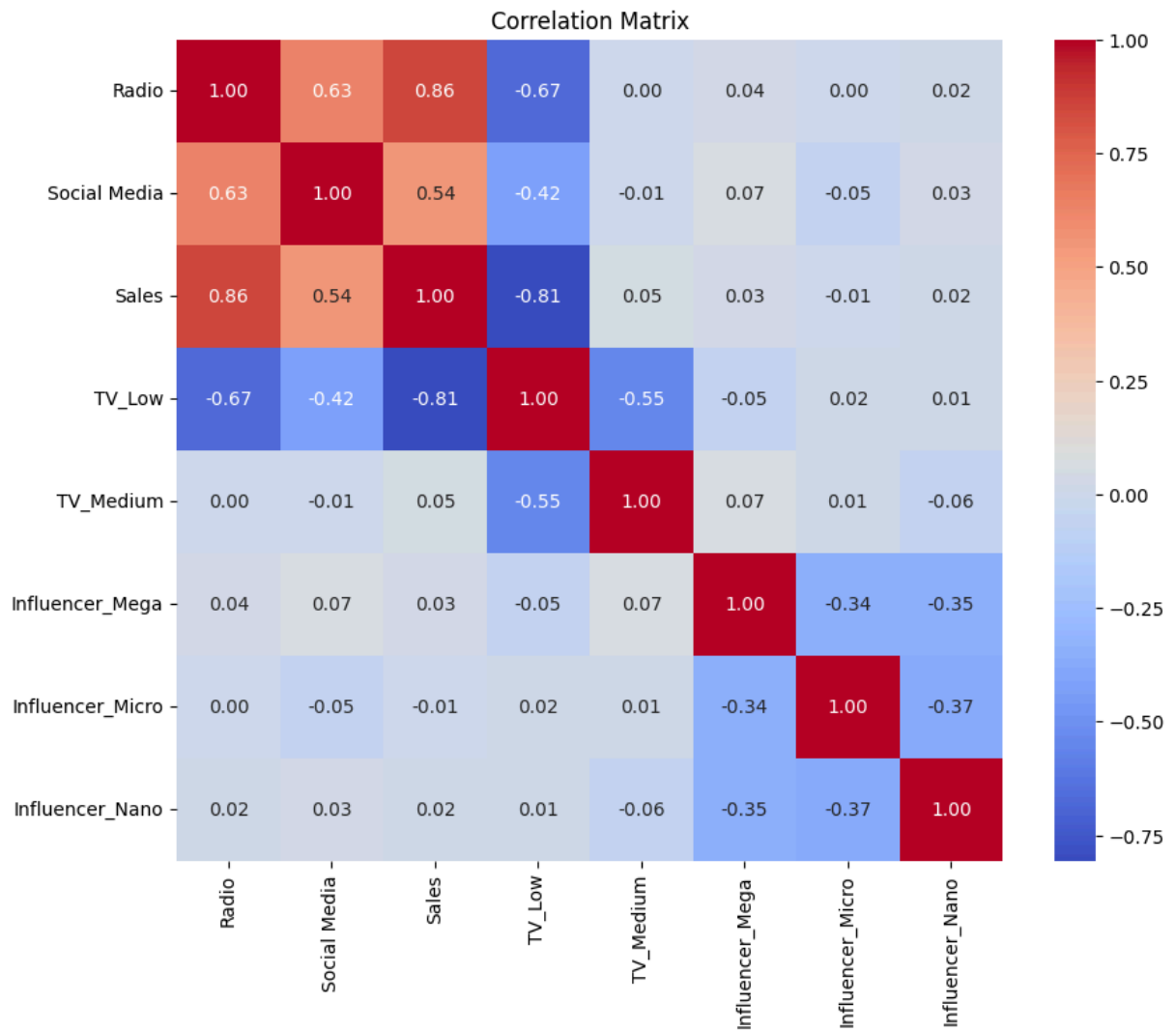


Correlation Matrix

```
In [48]: plt.figure(figsize=(10,8))

sns.heatmap(
    df.corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Matrix")
plt.show()
```



```
In [49]: # Define predictors and target
X = df.drop("Sales", axis=1)
y = df["Sales"]

# Convert predictors to numeric
X_numeric = X.astype(float)

print(X_numeric.dtypes)
print(X_numeric.values.dtype)
```

```
Radio          float64
Social Media   float64
TV_Low         float64
TV_Medium      float64
Influencer_Mega float64
Influencer_Micro float64
Influencer_Nano float64
dtype: object
float64
```

Multicollinearity Analysis

Variance Inflation Factor (VIF) was calculated for all predictors.

Radio (VIF = 6.70) and Social Media (VIF = 5.34) show moderate multicollinearity, which is consistent with the correlation matrix where both variables exhibited a positive correlation.

However, all VIF values remain below the commonly used threshold of 10, indicating that severe multicollinearity is not present. Therefore, all predictors were retained for the regression model.

```
In [50]: from statsmodels.stats.outliers_influence import variance_inflation_factor
import pandas as pd

X_numeric = X.astype(float)

vif_data = pd.DataFrame()
vif_data["Variable"] = X_numeric.columns

vif_data["VIF"] = [
    variance_inflation_factor(X_numeric.values, i)
    for i in range(X_numeric.shape[1])
]

print(vif_data)
```

	Variable	VIF
0	Radio	6.695980
1	Social Media	5.344600
2	TV_Low	1.661129
3	TV_Medium	1.664554
4	Influencer_Mega	1.996014
5	Influencer_Micro	2.083851
6	Influencer_Nano	2.084801

```
In [51]: # Add intercept
X = sm.add_constant(X_numeric)

# Fit model
model = sm.OLS(y, X).fit()

print(model.summary())
```

OLS Regression Results

Dep. Variable:	Sales	R-squared:	0.904			
Model:	OLS	Adj. R-squared:	0.903			
Method:	Least Squares	F-statistic:	760.4			
Date:	Tue, 23 Jun 2026	Prob (F-statistic):	1.82e-282			
Time:	13:22:14	Log-Likelihood:	-2713.4			
No. Observations:	572	AIC:	5443.			
Df Residuals:	564	BIC:	5478.			
Df Model:	7					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	217.4784	6.584	33.031	0.000	204.546	230.411
Radio	2.9735	0.235	12.644	0.000	2.512	3.435
Social Media	-0.1391	0.676	-0.206	0.837	-1.467	1.189
TV_Low	-154.5736	4.949	-31.231	0.000	-164.295	-144.852
TV_Medium	-75.5947	3.647	-20.726	0.000	-82.759	-68.431
Influencer_Mega	2.4948	3.462	0.721	0.471	-4.305	9.295
Influencer_Micro	2.9391	3.378	0.870	0.385	-3.695	9.574
Influencer_Nano	0.8015	3.346	0.240	0.811	-5.770	7.373
=====						
Omnibus:	58.711	Durbin-Watson:	1.874			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	17.802			
Skew:	0.057	Prob(JB):	0.000136			
Kurtosis:	2.143	Cond. No.	147.			
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

The model achieved an Adjusted R-squared of 0.903, indicating that approximately 90.3% of the variation in Sales can be explained by the selected marketing variables. The small difference between R-squared and Adjusted R-squared suggests that the included predictors contribute meaningful explanatory power. The overall regression model is statistically significant ($F = 760.4$, $p < 0.001$), indicating that the predictors collectively explain a significant portion of Sales variation.

Variable Inclusion / Exclusion Discussion

Radio, TV_Low, and TV_Medium were statistically significant predictors of Sales ($p < 0.05$).

Social Media and the Influencer category variables exhibited high p-values and therefore did not demonstrate statistically significant relationships with Sales after controlling for the other predictors.

Future model simplification could consider removing these non-significant variables and comparing Adjusted R-squared values to determine whether a more parsimonious model performs similarly.

Diagnostic Plots

```
In [52]: # Predicted values and residuals

predictions = model.fittedvalues
residuals = model.resid
```

Residual vs Fitted Plot

Linearity

The relationship between predictors and Sales was evaluated using a Residuals vs Fitted plot. Residuals appeared randomly distributed around zero, supporting the linearity assumption.

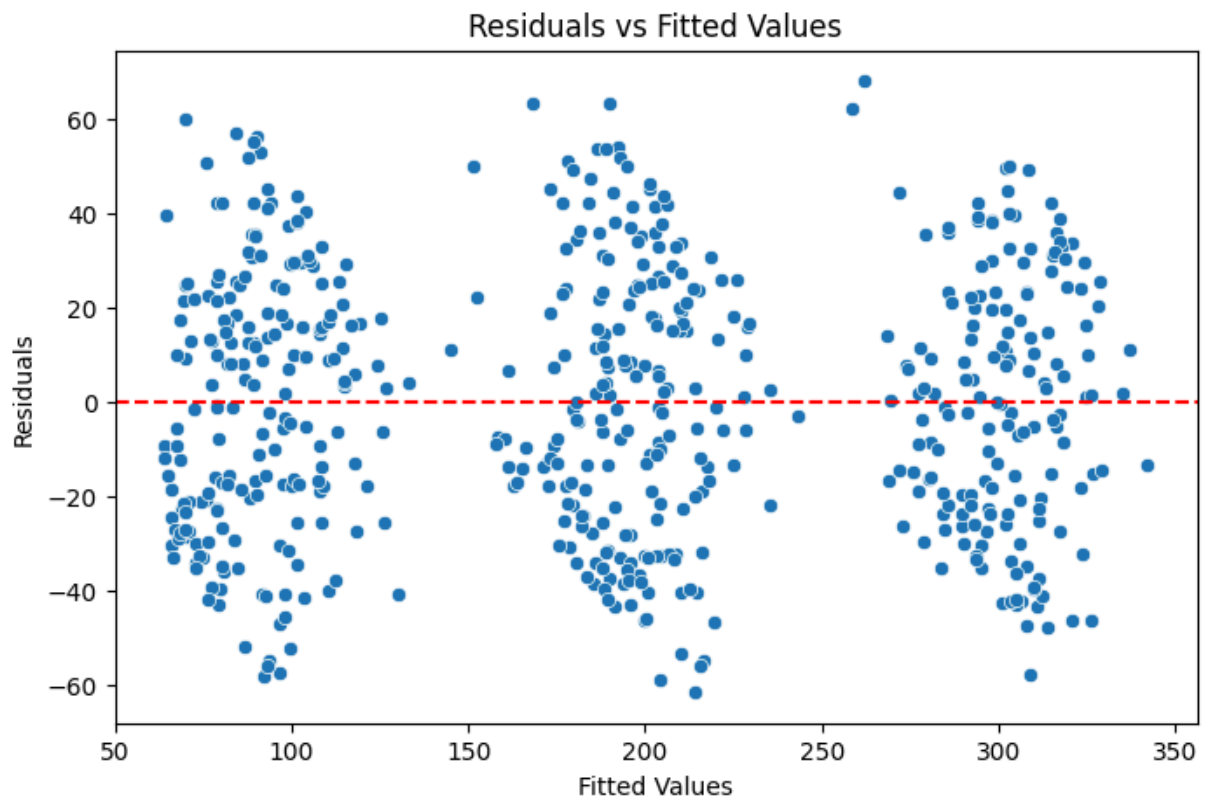
```
In [53]: plt.figure(figsize=(8,5))

sns.scatterplot(
    x=predictions,
    y=residuals
)

plt.axhline(0, color='red', linestyle='--')

plt.xlabel("Fitted Values")
plt.ylabel("Residuals")
plt.title("Residuals vs Fitted Values")

plt.show()
```



Q-Q Plot (Normality)

Normality

A histogram of residuals and a Q-Q plot were examined. Residuals were approximately normally distributed, with minor deviations in the tails. Given the large sample size ($n = 572$), these deviations are unlikely to materially affect model validity.

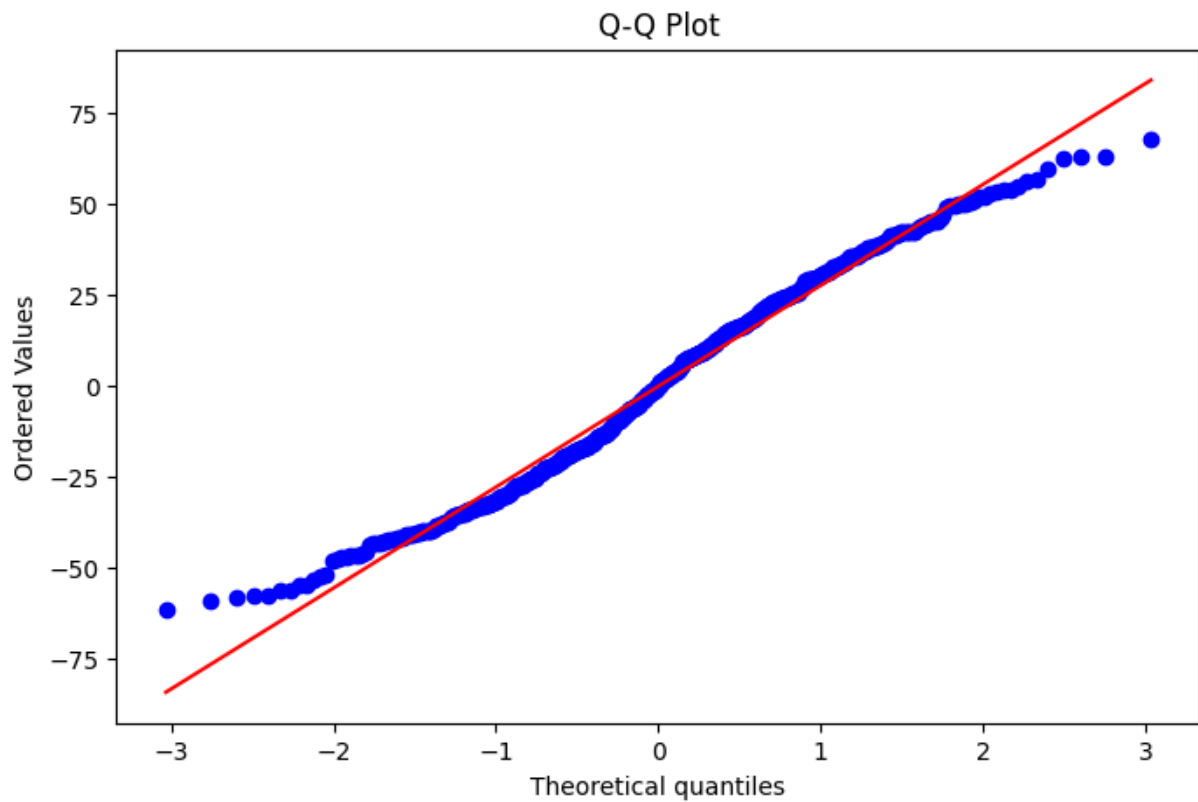
```
In [54]: import scipy.stats as stats

plt.figure(figsize=(8,5))

stats.probplot(
    residuals,
    dist="norm",
    plot=plt
)

plt.title("Q-Q Plot")

plt.show()
```



Histogram of Residuals

Homoscedasticity

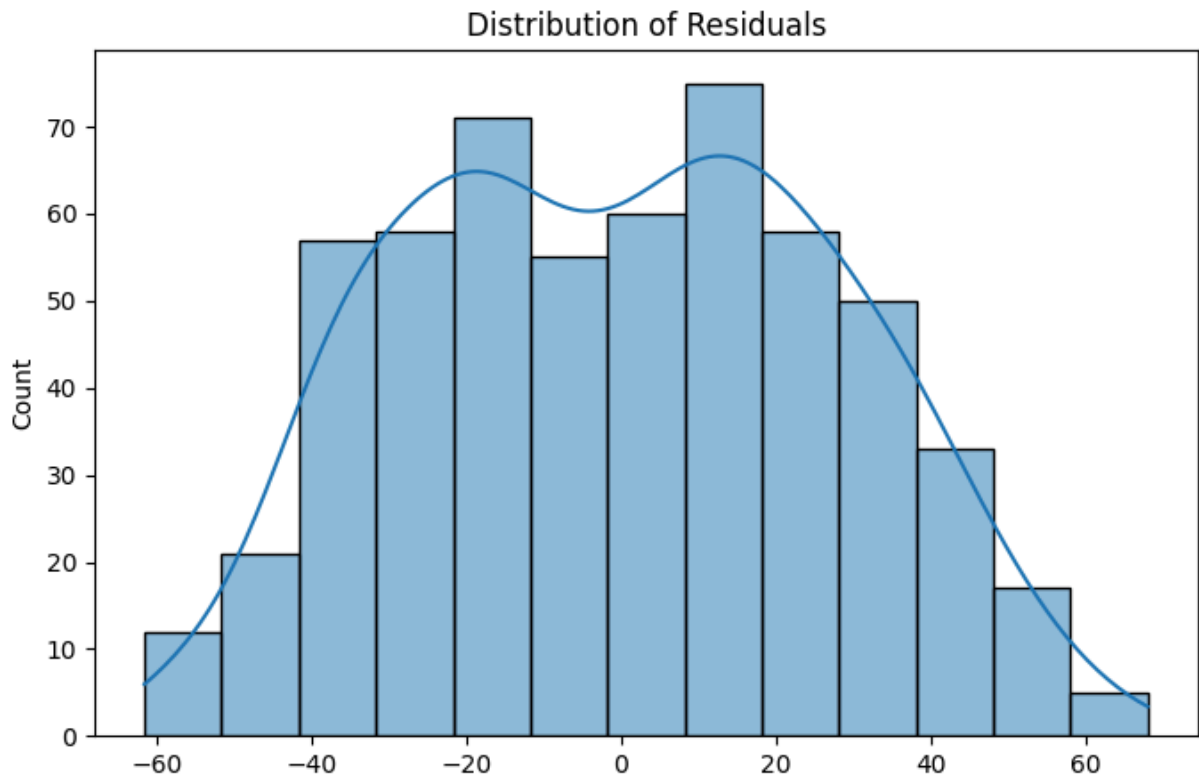
Residuals exhibited relatively constant variance across fitted values, suggesting that the homoscedasticity assumption was reasonably satisfied.

```
In [55]: plt.figure(figsize=(8,5))

sns.histplot(
    residuals,
    kde=True
)

plt.title("Distribution of Residuals")

plt.show()
```



Business Recommendation

The multiple linear regression model explained approximately 90.3% of the variation in Sales (Adjusted $R^2 = 0.903$), indicating excellent predictive performance.

Radio advertising emerged as the strongest significant positive predictor of Sales. Holding all other variables constant, each one-unit increase in Radio spend was associated with an average increase of approximately 2.97 units in Sales.

TV advertising level also significantly influenced Sales. Relative to the reference TV category, campaigns classified as TV_Low and TV_Medium were associated with substantially lower Sales outcomes, suggesting that reduced TV investment may negatively affect performance.

Social Media spend and Influencer categories did not demonstrate statistically significant effects after accounting for other variables in the model.

Based on these findings, marketing budget allocation should prioritize Radio advertising and maintain strong TV investment levels. Resources devoted to Social Media and Influencer campaigns should be reviewed and further tested before receiving additional budget increases.

In []: